

MagicCode Super-Resolution Software Migration Guide

Document Version: v2.2.0. Release Date: September 30, 2026.

1. Purpose

This guide helps customers migrate to MagicCode Super-Resolution Software v2.2.0, including from v2.0.0/v2.1.0 and from v1.2.x spatial-only public docs.

2. Main changes

Area	v1.2.x / v2.0.x / v2.1.0	v2.2.0	Action
Version	v1.2.0 / v2.0.0 / v2.1.0	v2.2.0	Rebuild against new headers/libs.
Modes	SPEED_MODE / BALANCED_MODE (spatial)	SPATIAL_* plus TEMPORAL_SPEED_MODE and TEMPORAL_BALANCED_MODE	Rename spatial enumerators; add 2/3 only for temporal.
MC_Process	Some docs showed (handle, in, out)	mc_nscaler_enable(&handle, &in_frame, &out_frame)	Split in/out frames; temporal needs frame.
Scale	[1,8] for documented spatial	Spatial [1,8]; temporal (1,8]	Reject temporal 1.0.
Windows	OpenGL often documented alone	Vulkan / D3D11 / OpenGL 4.3 temporal on session API	Enable default remains OpenGL; D3D11 is session-only.
Enable helper	v2.1.0 MC_Enable spatial helper	Removed in v2.2.0	Default-create via mc_nscaler_enable (NULL handle).
Public API	MC_Init / MC_Process / MC_Uninit / MC_Control	mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable / mc_nscaler_version	Rename calls; create via SET_PARAM or NULL-handle enable.
Create params	input_param_t (struct_size, width, height)	ctrl_param_t; size on mc_nscaler_input_frame_t	Drop struct_size; move width/height to the frame.
Frame struct	magic_frame_t (image_in/image_out)	mc_nscaler_input_frame_t + mc_nscaler_output_frame_t	Split input/output; temporal_frame_t stays on .frame.
Platforms	Android, iOS/iPadOS/macOS, Windows	+ Linux x86_64 (CPU spatial archive)	Link lib/linux/libmagic_sr.a.

Temporal Super-Resolution

Temporal super-resolution runs through the nscaler session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable. v2.2.0 replaced the v2.0.0 MC_Init / MC_Process / MC_Uninit entry and removed the v2.1.0 MC_Enable helper (mc_enable.h is deleted). A NULL *handle on mc_nscaler_enable creates a session with defaults (2x SPATIAL_SPEED) and processes the first frame.

Modes

Value	Enumerator	Role
0	SPATIAL_SPEED_MODE	Spatial, throughput first
1	SPATIAL_BALANCED_MODE	Spatial, quality/cost trade-off
2	TEMPORAL_SPEED_MODE	Temporal, throughput first
3	TEMPORAL_BALANCED_MODE	Temporal, canonical FSR-family path

- Temporal scaler_factor is (1, 8]; 1.0 itself is rejected. Spatial remains [1.0, 8.0].
- TEMPORAL_SPEED_MODE on Metal, Vulkan, and OpenGLS currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family temporal path (same family as Balanced, not a separate customer API).
- TEMPORAL_BALANCED_MODE uses the canonical FSR-family path on every GPU temporal backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLS do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run temporal modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- mc_nscaler_input_frame_t / mc_nscaler_output_frame_t: caller-owned color and output GPU resources.
- mc_nscaler_input_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at create (mc_nscaler_control SET_PARAM, or the first mc_nscaler_enable): Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan temporal requires mc_nscaler_input_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.

- Create-stage depth_reversed / depth_infinite / hdr_color on ctrl_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both temporal modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

3. Native code migration

1. Replace headers with v2.2.0 mc_interface.h (mc_enable.h is removed).
2. Rebuild against v2.2.0 libraries (one archive per platform: libmagic_sr.a / libmagic_sr.lib; Linux: lib/linux/libmagic_sr.a).
3. Set struct_size on temporal_frame_t (ctrl_param_t has no struct_size; the image size moves to mc_nscaler_input_frame_t).
4. Replace MC_Init / MC_Process / MC_Uninit and MC_Enable calls with mc_nscaler_control(SET_PARAM) create, mc_nscaler_enable per frame, and mc_nscaler_disable at shutdown.
5. For temporal, import gpu_context and pass depth, motion, jitter, camera, and (Vulkan) a recording command_buffer.
6. Use QUERY_STATUS; inspect temporal_* fields only when temporal_status_valid is set.

4. Unity / Unreal

Replace plugin binaries as you republish. Plugin spatial APIs do not feed temporal extras. Native v2.2.0 temporal is the nscaler session API (mc_nscaler_enable with temporal_frame_t).

5. Documentation migration

- Use User_Guide.md / 用户使用说明书.md v2.2.0 for customer integration.
- Legal documents (EULA, Privacy Policy, Data Reporting Notice) keep their existing substance and are not version-bumped by this technical drop.
- Filenames under doc/版本文档 no longer include a version suffix.

6. Acceptance checklist

- mc_nscaler_version returns v2.2.0.
- Spatial processing works on target mobile backends.
- If shipping temporal: TEMPORAL_SPEED_MODE and TEMPORAL_BALANCED_MODE process with temporal_frame_t depth/motion/jitter.
- CPU temporal is rejected as expected.
- Linux x86_64 archive links and runs CPU spatial (AVX2).
- Known Issues reviewed.